

EduMind AI

Smart AI-Powered Learning Assistant

Project Presentation

Project Overview

EduMind AI is an AI-powered educational learning assistant.

Students can ask questions and receive AI-generated answers and explanations.

The platform combines multiple learning tools in one interface.

Google Gemini provides the AI-powered content generation.

The interface is designed for simple, interactive learning.

Project Objectives

Provide an interactive way for students to ask learning questions.

Explain technical and academic concepts in a beginner-friendly format.

Support practice through AI-generated quizzes.

Create structured study notes and summaries.

Recommend a learning path based on topic and learner level.

Key Features

AI Q&A

Ask questions on programming and learning topics.

Explain

Get simplified concept explanations.

Quiz

Generate practice questions for a topic.

Study Notes

Create structured revision notes.

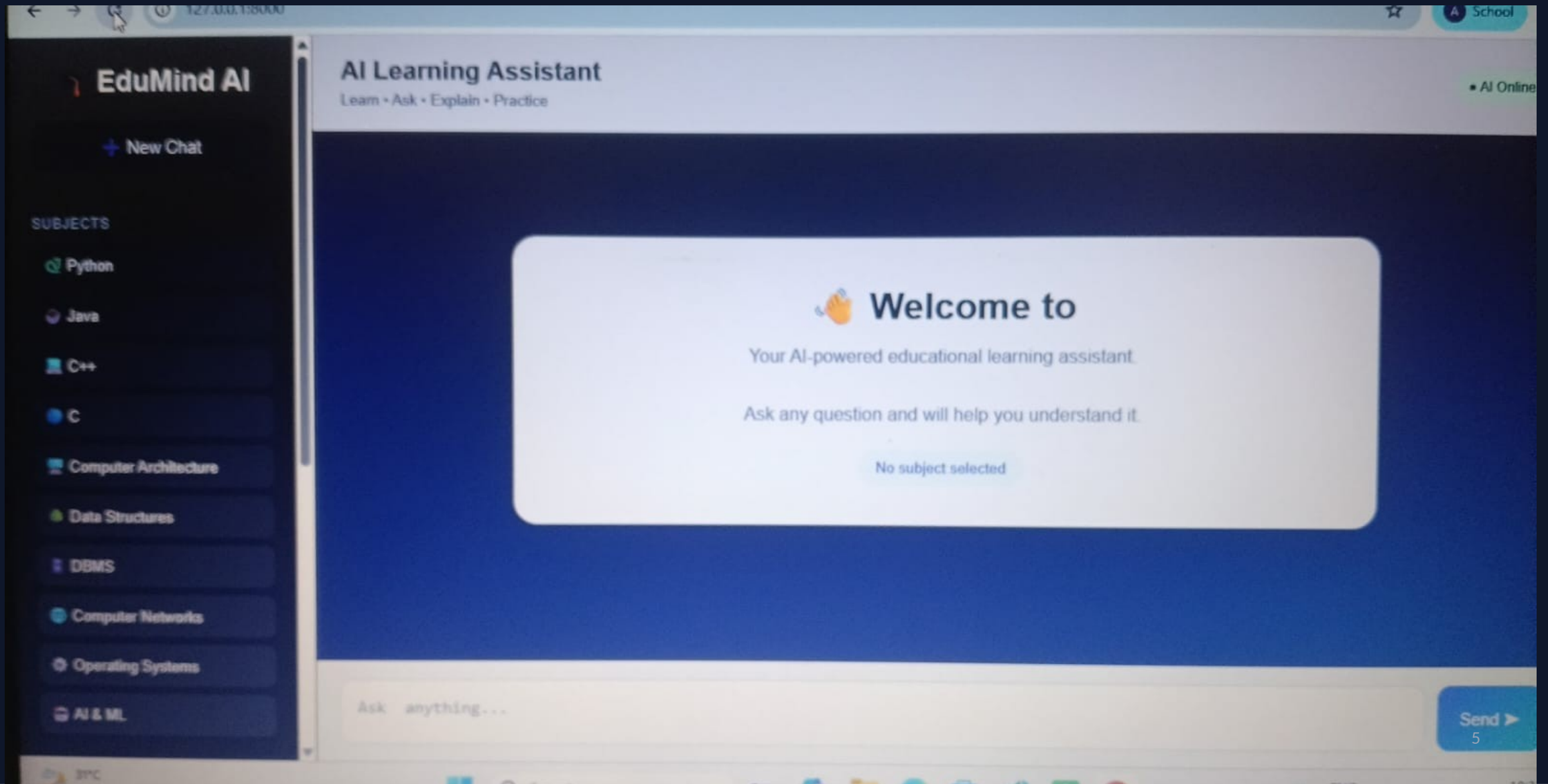
Summarize

Turn longer content into concise summaries.

Learning Path

Get a structured learning path.

EduMind AI User Interface



Technology Stack

Frontend: HTML, CSS and JavaScript

Backend: Python and FastAPI

AI Integration: Google Gemini API

Templates: Jinja2

Configuration: environment variables (.env)

Development Environment: Visual Studio Code

Application Flow

1. Student selects a subject or learning tool.
2. Student enters a question or topic.
3. FastAPI receives the request.
4. The relevant Python module prepares the AI prompt.
5. Google Gemini generates the response.
6. EduMind AI displays the result in the web interface.

Backend Modules

main.py — FastAPI application and API routes

gemini_client.py — Gemini API communication

qna.py — question answering

explanation_module.py — concept explanations

quiz_module.py — quiz generation

notes_module.py — study notes

summary_module.py — summarization

learning_path.py — learning path recommendations

Project Demonstration

Select Python from the subject list.

Ask: “What is a variable in Python?”

Show the AI-generated answer.

Demonstrate Explain, Quiz, Study Notes, Summarize and Learning Path.

Show how one platform supports different learning activities.

Future Enhancements

Voice-based questions and responses.

Student progress tracking and dashboards.

Multilingual learning support.

Personalized learning recommendations.

More interactive practice and assessment features.

Deployment as a production-ready web application.

Thank You

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